CODEX_BRIDGE.md — CODEX↔ABACUS Bridge Architecture

> **Version:** 1.0.0
> **Purpose:** Documents the relationship and data flow between GBOGEB/CODEX and GBOGEB/ABACUS

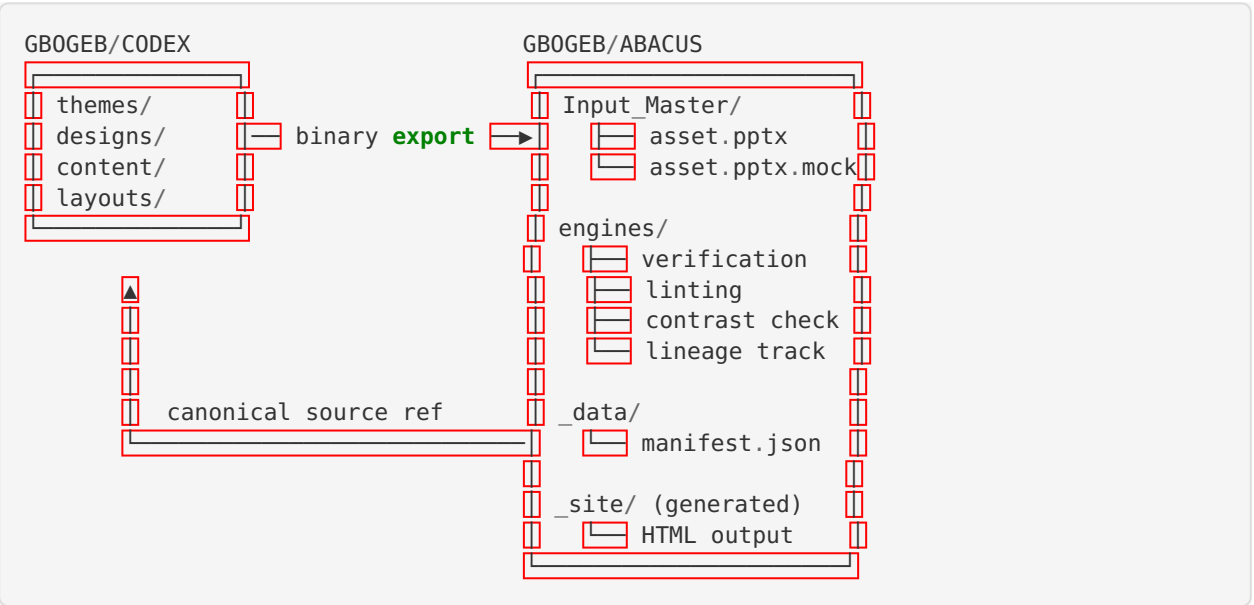
Overview

The GBOGEB system operates as a **two-repository architecture**:

Repository	Role	Contents
GBOGEB/CODEX	Design & Theme Blueprint	Visual themes, interface designs, canonical source content
GBOGEB/ABACUS	Governance & Processing Engine	Validation, lineage tracking, YAML compliance, rendering

The bridge between them operates through the `Input_Master/` folder with `.mock` sidecar files providing immutable metadata.

Data Flow



Bridge Protocol

Step 1: Asset Export (CODEX → ABACUS)

1. Designer creates or updates content in CODEX
2. Binary exports (PPTX, PDF, images) are placed in `Input_Master/`
3. No manual editing of exported files is permitted

Step 2: Verification (ABACUS Input Gate)

1. `verification_hook.py` processes each binary file
2. SHA256 hash computed and stored in `.mock` sidecar
3. `lineage_manifest.json` updated with asset metadata
4. `.mock` files are committed; binaries are `.gitignored`

Step 3: Governance Validation (ABACUS Engines)

1. `RENDER_LINTER.py` validates content structure
2. `WCAG_CONTRAST_CHECKER.py` ensures accessibility compliance
3. `SLIDE_ID_ENFORCER.py` verifies deterministic identifiers
4. All results must pass before rendering proceeds

Step 4: Rendering (ABACUS Output)

1. Jekyll processes validated content with semantic themes
2. HTML-first output generated using `_layouts/` templates
3. Lineage records created linking output to source
4. Generated outputs carry lineage metadata

Step 5: Lineage Closure

1. Every output is traceable back to its CODEX source
2. `derived_from` field points to CODEX repository path
3. `render_commit` pins the exact ABACUS engine state

Core Principle

> **“Generated outputs are NEVER canonical.”**

This means:
- PPT files exported from CODEX are **source artifacts**, not editable outputs
- HTML rendered by ABACUS is a **derived output**, not a source of truth
- Any modification to generated files creates a **lineage violation**
- The only way to change output is to update CODEX source and re-render

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Cross-Repository Dependencies

ABACUS Component	CODEX Dependency	Sync Method
`SEMANTIC_THEME.yaml`	`themes/`	Manual sync on theme update
`LAYOUT_CONTRACTS.yaml`	`layouts/`	Version-pinned reference
`Input_Master/*.mock`	Binary exports	Verification hook
`lineage_manifest.json`	All assets	Automated tracking

Bridge Integrity Checks

- **Hash verification**: SHA256 of Input_Master/ files must match .mock records
- **Manifest completeness**: Every .mock file must have a manifest entry
- **Lineage chain**: Every output must trace to a CODEX source path
- **Version pinning**: ABACUS render_commit is recorded for reproducibility